

# Phase 0 (Spike) — Work Breakdown: phases → tasks → subtasks

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**Revision: 2 Last modified: 2026-07-04T12:00:00Z Rev 2:**  
Independent gap-analysis pass (§14 risk register now cross-references v00-meta/decision-register.md's per-decision reversal criteria so a gate failure's architectural consequence is traceable to the exact decision it re-opens, not just a prose fallback). No contradictions found against HelixVPN-Phase0-Spike.md (the original CLD source) or v07-execution/{workable-items-model, dependency-graph, subtask-deepening-p1}.md — this WBS's 3-tier epic→task→subtask breakdown, gate evidence methodology (§12), and risk register (§14) were independently verified already concrete (falsifiable acceptance + captured-evidence method per item); R5 (task/subtask tier asymmetry, REFINEMENT\_NOTES.md) does not apply to this document — Phase 0 was always the 3-tier side of the asymmetry.

Master technical specification — document 06 of the HelixVPN set. Scope: the **complete Work Breakdown Structure (WBS) for Phase 0 — the Spike**: every exit gate G1..G6 (question + go/no-go bar), every milestone S0..S8 decomposed into numbered **TASKS** and **SUBTASKS** with stable DB-ready ids HVPN-P0-NNN, and the surviving interfaces / harnesses (iOS memory rig, Go-vs-Rust edge benchmark, netns + nftables-DPI + netem rig with concrete commands). This is a SPEC — it describes the work and its interfaces; it does NOT build the product (2–3 refinement passes follow). This document is the **single source for the workable-items SQLite DB** (§11.4.93/.95): the HVPN-P0-NNN rows + fields below are authored to load directly.

Source evidence cited inline by id: [04\_P0] = 04\_VPN\_CLD/  
HelixVPN-Phase0-Spike.md; [04\_ARCH] = 04\_VPN\_CLD/HelixVPN-  
Architecture-Refined.md; [04\_P1] = Phase1-MVP;  
[SYNTHESIS]; LLM analyses [02\_QWN] [01\_DSK] [07\_GMI]  
[10\_KMI] [11\_MST]; external facts [research-masque] (RFC  
9297/9298/9221), [research-boringtun], [research-quinn].

## 0. How to read this WBS (and the DB schema it populates)

Phase 0 is **time-boxed to ~3-4 focused weeks** [04\_P0 §0]. It produces throwaway-quality *implementations* behind production-quality *interfaces*: the traits, FFI signatures, and wire contracts here are meant to **survive into Phase 1** [04\_P0 intro]. A gate that cannot be cleared inside the box **is the finding** — escalate the decision rather than overrun silently [04\_P0 §0, §13].

### 0.1 Decomposition hierarchy

|                  |                  |                                                  |
|------------------|------------------|--------------------------------------------------|
| PHASE 0 (Spike)  |                  |                                                  |
| ├─ EXIT GATES    | G1..G6           | – the go/no-go questions (§1)                    |
| ├─ MILESTONES    | S0..S8           | – independently demoable slices (§3..§11)        |
| │ └─ TASKS       | HVPN-P0-NNN      | – a unit of delivery (own acceptance + evidence) |
| │ └─ SUBTASKS    | HVPN-P0-NNN      | – the smallest tracked work, each its own row    |
| └─ CROSS-CUTTING | HVPN-P0-077..080 | – harness, decision log, DB/CI, infra (§12)      |

Every TASK and every SUBTASK is a **workable item** with the same closed field set, so the table rows below load 1:1 into the docs/workable\_items.db schema (§0.3). IDs are **monotonic, never renumbered** (§11.4.54): allocated once, parent expresses hierarchy.

### 0.2 Field dictionary (every item row carries these)

| Field  | Meaning                                                                  |
|--------|--------------------------------------------------------------------------|
| id     | stable HVPN-P0-NNN (§11.4.54) — primary key, never reused                |
| parent | the owning milestone id (S0..S8) or task id; – for milestone-level tasks |
| kind   | task   subtask   gate                                                    |
| title  | ≥6-word self-contained meaning (§11.4.91)                                |

| Field       | Meaning                                                                       |
|-------------|-------------------------------------------------------------------------------|
| gate        | which exit gate(s) this item advances: G1..G6 or –                            |
| description | what / how it manifests / acceptance intent (§11.4.148 D2)                    |
| depends_on  | upstream HVPN-P0-NNN ids that MUST be done first                              |
| deliverable | the concrete artifact produced (file/binary/CSV/report)                       |
| acceptance  | falsifiable pass condition with captured PHYSICAL evidence (§11.4.5/.69/.107) |
| est_effort  | engineer-days (rough, [04_P0 §2] day estimates are the anchor)                |
| test_types  | the §11.4.169 closed-set test-type codes required (§0.4)                      |

### 0.3 Workable-items DDL (§11.4.93 — this WBS loads into it)

-- docs/workable\_items.db (git-tracked, §11.4.95). Phase-0 subset of the canonical schema.

```
CREATE TABLE IF NOT EXISTS items (
  id          TEXT PRIMARY KEY,          --
    'HVPN-P0-001'
  parent      TEXT REFERENCES items(id), --
    milestone/task ownership
  kind        TEXT NOT NULL CHECK (kind IN
    ('task','subtask','gate','milestone')),
  type        TEXT NOT NULL CHECK (type IN
    ('Bug','Feature','Task')) DEFAULT 'Task',
  status      TEXT NOT NULL DEFAULT 'Queued', --
    §11.4.15 closed set
  severity    TEXT NOT NULL DEFAULT 'Normal',
  title       TEXT NOT NULL CHECK (length(title) >= 40), --
    §11.4.91
  gate        TEXT,                      --
    'G1'..'G6' or NULL
  description TEXT NOT NULL CHECK (length(description) >= 40),
  depends_on  TEXT NOT NULL DEFAULT '[]', -- JSON
    array of ids
  deliverable TEXT NOT NULL,
  acceptance  TEXT NOT NULL,
  est_effort_d REAL NOT NULL DEFAULT 1.0,
  test_types  TEXT NOT NULL DEFAULT '[]', -- JSON
    array of §11.4.169 codes
  created_at  TEXT NOT NULL,
  modified_at TEXT NOT NULL
);
CREATE TABLE IF NOT EXISTS gates (
  id          TEXT PRIMARY KEY,          --
    'G1'..'G6'
  question    TEXT NOT NULL,
```

```

go_no_go_bar TEXT NOT NULL,
section      TEXT NOT NULL,
outcome      TEXT NOT NULL DEFAULT 'pending'
             CHECK (outcome IN
                   ('pending', 'pass', 'fail', 'rust', 'go')),
evidence_path TEXT --
             §11.4.5 captured evidence
);
CREATE INDEX IF NOT EXISTS idx_items_parent ON items(parent);
CREATE INDEX IF NOT EXISTS idx_items_gate  ON items(gate);

```

## 0.4 §11.4.169 test-type code map (used in the test\_types column)

| Code | §11.4.169 test type                                                | Phase-0 instantiation                              |
|------|--------------------------------------------------------------------|----------------------------------------------------|
| UT   | unit                                                               | Rust #[cfg(test)], Dart test, Go _test.go          |
| IT   | integration (real System, infra via containers submodule §11.4.76) | netns rig + edge + connector, booted on-demand     |
| E2E  | end-to-end                                                         | client overlay namespace → curl http://10.10.0.20/ |
| FA   | full-automation (§11.4.25/.52/.98, deterministic §11.4.50)         | make spike one-shot, N=3 identical runs            |
| CH   | Challenges (challenges submodule §11.4.27(B))                      | per-gate Challenge scoring on captured evidence    |
| HQA  | HelixQA (helix_qa submodule)                                       | autonomous QA session against the slice            |
| LOAD | DDoS / load-flood                                                  | 1/10/100 concurrent iperf3 + handshake-flood       |
| SEC  | security (§11.4.10 + security submodule)                           | DPI evasion, wire fingerprint, no-plaintext-leak   |
| SC   | stress + chaos (§11.4.85)                                          | iface flap, packet loss, process kill mid-transfer |
| CONC | concurrency / atomicity                                            | three-loop orchestrator under concurrent send/recv |
| RACE | race-condition / deadlock                                          | cargo test --features loom, -race on Go edge       |
| MEM  | memory                                                             | RSS sampling — iOS Instruments / /proc / dumpsys   |

| Code  | §11.4.169 test type        | Phase-0 instantiation                              |
|-------|----------------------------|----------------------------------------------------|
| BENCH | benchmarking / performance | bench.sh throughput /<br>CPU-per-Gbps / p99<br>CSV |

Per §11.4.169, the ONLY permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason (e.g. iOS MEM on a machine with no device → topology\_unsupported), never a silent gap. Four-layer enforcement per §11.4.4(b) applies to every TASK closure.

## 1. Exit gates G1..G6 (the questions Phase 0 must answer)

A "no-go" on **G3** or **G4** changes the architecture, not just the schedule — that is exactly why they are in Phase 0 [04\_P0 §0]. Gates load into the gates table (§0.3).

| Gate      | Question                                                                                                   | Go / No-go bar                                                                                                                                                                              |
|-----------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>G1</b> | Can helix-core (Rust + boringtun) move real traffic client→gateway→connector LAN over <b>plain UDP</b> ?   | iperf3 through-tunnel $\geq$ <b>80%</b> of bare-link throughput; ping/curl to 10.10.0.20:80 succeed                                                                                         |
| <b>G2</b> | Can the <b>same core</b> do it over <b>MASQUE/QUIC</b> (WG-in-HTTP/3)?                                     | Works through a DPI-style UDP block on :51820; $\geq$ <b>50%</b> of plain-UDP throughput; survives 5% loss better than a UDP-over-TCP strawman; flow classified HTTP/3 with no WG signature |
| <b>G3</b> | Does the Rust core run inside an <b>iOS NEPacketTunnelProvider</b> under its memory ceiling with headroom? | Extension-process steady-state RSS under the device-enforced ceiling with $\geq$ <b>30% headroom</b> across a 30-min, 1 GB transfer, for <b>both</b> plain UDP and MASQUE                   |
| <b>G4</b> | <b>Go vs Rust for the gateway edge</b> — which terminates MASQUE better, all-in?                           | Decision <b>recorded</b> with benchmark numbers (CPU/Gbps, p99, churn, memory) + reuse/velocity assessment                                                                                  |
| <b>G5</b> | Is the <b>flutter_rust_bridge FFI</b> boundary clean enough to drive the core from Dart?                   | Flutter-Linux app toggles the tunnel and shows live status (Connecting→Handshaking→Connected{transport, r...} driven only by the core's event stream                                        |
| <b>G6</b> | Does <b>push-based reconciliation</b> work at all (static map → core brings up peers)?                     | Core consumes a map.json delta and converges <b>without restart</b> ; the newly-advertised network becomes reachable                                                                        |

```

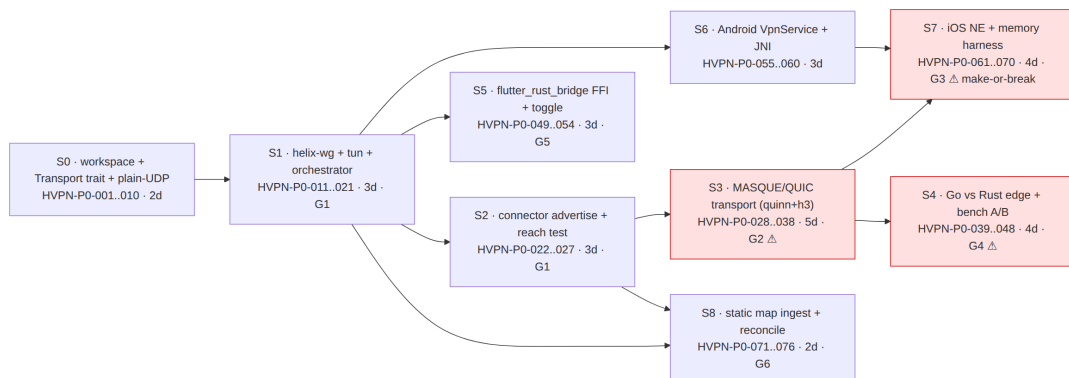
INSERT INTO gates (id,question,go_no_go_bar,section,outcome)
VALUES
('G1','plain-UDP slice client→gw→connector
      LAN','iperf3 ≥80% bare link; curl 10.10.0.20:80 OK','§3-
      §5','pending'),
('G2','same core over MASQUE/QUIC through DPI block','≥50% of
      plain-UDP; survives 5% loss > UoT; classified HTTP/
      3','§6','pending'),
('G3','Rust core in iOS NEPacketTunnelProvider memory','steady
      RSS < ceiling with ≥30% headroom, both transports, 30-min
      1GB','§10','pending'),
('G4','Go vs Rust gateway edge for MASQUE','decision recorded
      with CPU/Gbps, p99, churn, mem +
      velocity','§7','pending'),
('G5','flutter_rust_bridge FFI drives core from Dart','Flutter-
      Linux toggle + live status from event
      stream','§8','pending'),
('G6','push-based reconcile from static map','map delta →
      converge, no restart, network reachable','§11','pending');

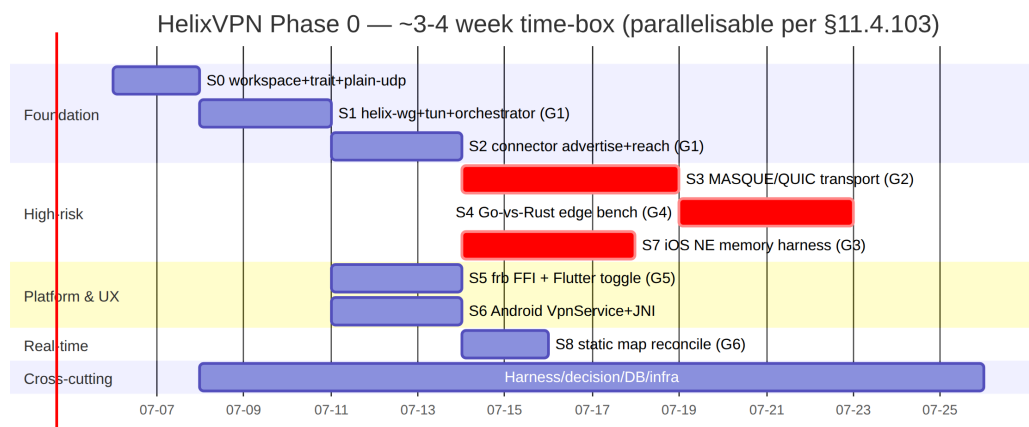
```

---

## 2. Milestone map (dependency graph + schedule)

S3, S4, S7 are the high-risk milestones; sequence so a failure surfaces early. S0–S2 are the foundation everything else stands on [04\_P0 §2].





**Parallelism note (§11.4.103/.94):** S5/S6/S8 do not depend on S3/S4 and run as background streams concurrently with the high-risk critical path; S4 and S7 must **not** share the single benchmark host at once (§11.4.119 single-resource-owner — the bench rig is an exclusive resource; partition device ownership).

### 3. Milestone S0 — Cargo workspace + Transport trait + plain-UDP (HVPN-P0-001..010)

**Proves:** the abstraction compiles and round-trips. **Effort:** ~2d. **Gate:** foundation for G1. **Demo:** loopback echo test green.

#### TASK HPVNP0-001 — Bootstrap the helix-core Cargo workspace skeleton

| Field       | Value                                                                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S0                                                                                                                                                                                                                                                                     |
| kind        | task · type Task                                                                                                                                                                                                                                                       |
| title       | Bootstrap helix-core Cargo workspace with the six surviving crates                                                                                                                                                                                                     |
| gate        | — (foundation for G1/G5)                                                                                                                                                                                                                                               |
| description | Create the [workspace] with crates/{helix-transport,helix-wg,helix-tun,helix-core,helix-ffi} + bin/{helix-client,helix-connector} per [04_P0 §4.1]; pin Rust toolchain (rust-toolchain.toml, 1.82+); wire CI-less local check (cargo check --workspace) per §11.4.156. |
| depends_on  | —                                                                                                                                                                                                                                                                      |
| deliverable |                                                                                                                                                                                                                                                                        |

| Field      | Value                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------------|
|            | helix-core/Cargo.toml + crate stubs that cargo check --workspace passes                                     |
| acceptance | cargo check --workspace exit 0; cargo tree shows the 6-crate graph; captured terminal recording (§11.4.158) |
| est_effort | 0.5                                                                                                         |
| test_types | UT                                                                                                          |

Surviving workspace layout (must match Phase 1 monorepo helix-core/ [04\_ARCH §11]):

```

helix-core/
├─ Cargo.toml                # [workspace]
├─ rust-toolchain.toml       # channel = "1.82.0"
├─ crates/
│   └─ helix-transport/      # Transport trait + registry
(shared client+edge+connector)
│   │   └─ src/lib.rs
│   │   └─ src/plain_udp.rs  # S0/S1
│   │   └─ src/masque.rs     # S3
│   └─ helix-wg/             # boringtun wrapper
│   └─ helix-tun/            # OS tun abstraction (Linux now)
│   └─ helix-core/           # orchestrator: tun<->wg<-
>transport + event stream
│   └─ helix-ffi/            # flutter_rust_bridge surface (S5)
└─ bin/
    └─ helix-client.rs
    └─ helix-connector.rs

```

| Subtask                                                                   | id          | deliverable                            | acceptance                                         | test_types |
|---------------------------------------------------------------------------|-------------|----------------------------------------|----------------------------------------------------|------------|
| Workspace manifest + 6 crate stubs                                        | HVPN-P0-002 | Cargo.toml + lib.rs stubs              | cargo check --workspace exit 0                     | UT         |
| Toolchain pin + dep baseline (tokio, bytes, async-trait, anyhow, tracing) | HVPN-P0-003 | rust-toolchain.toml, locked Cargo.lock | cargo build --workspace exit 0; lockfile committed | UT         |

## TASK HVPN-P0-004 — Define the Transport trait + registry + dial() (THE survivor interface)

The single most important interface in the project [04\_P0 §4.2]: **one trait, three consumers, N transports**. Everything obfuscation-related hides behind it. The transport NEVER sees plaintext (invariant I1, [01-data-plane §0.1]).

| Field       | Value                                                                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S0                                                                                                                                                                                                                       |
| kind        | task · type Feature                                                                                                                                                                                                      |
| title       | Define the surviving Transport trait, TransportConfig, and dial() registry                                                                                                                                               |
| gate        | — (underpins G1, G2, G4)                                                                                                                                                                                                 |
| description | Author the byte-stable Transport trait carrying <b>unreliable WG datagrams</b> (I2) + TransportConfig enum + dial() factory that the auto-escalation ladder later drives. Interface MUST survive into Phase 1 unchanged. |
| depends_on  | HVPN-P0-002                                                                                                                                                                                                              |
| deliverable | helix-transport/src/lib.rs with the trait + config + dial()                                                                                                                                                              |
| acceptance  | trait object Box<dyn Transport> usable from client/connector/edge crates; doctests compile; §11.4.92 multi-pass review marker recorded                                                                                   |
| est_effort  | 0.5                                                                                                                                                                                                                      |
| test_types  | UT, CONC                                                                                                                                                                                                                 |

```
// helix-transport/src/lib.rs – the survivor interface [04_P0
    §4.2]
use async_trait::async_trait;
use bytes::Bytes;
use std::net::SocketAddr;

/// A bidirectional carrier for already-encrypted WireGuard
    datagrams.
/// The transport NEVER sees plaintext; it only changes how WG
    bytes look on the wire.
#[async_trait]
pub trait Transport: Send + Sync {
    async fn send(&self, datagram: Bytes) -> Result<(),
        TransportError>;    // one WG datagram out
    async fn recv(&self) -> Result<Bytes,
        TransportError>;    // next WG datagram in
        (cancel-safe)
    fn kind(&self) -> &'static str;    // "plain-udp"
        | "masque-h3" | ...
```

```

    fn effective_mtu(&self) ->
        u16; // WG MTU
        after transport overhead
}

#[derive(Clone, Debug)]
pub enum TransportConfig {
    PlainUdp { peer: SocketAddr, bind: SocketAddr },
    Masque { url: String, sni: String, bind: SocketAddr }, //
        https://gw:443
    // Phase 1+: Shadowsocks { .. }, UdpOverTcp { .. }, Lwo { .. }
}

#[derive(thiserror::Error, Debug)]
pub enum TransportError {
    #[error("io: {0}")] Io(#[from] std::io::Error),
    #[error("handshake failed: {0}")] Handshake(String),
    #[error("closed")] Closed,
}

/// Build a transport from config. The escalation ladder (auto
    mode) just
/// constructs the next variant on repeated handshake failure.
pub async fn dial(cfg: TransportConfig) -> Result<Box<dyn
    Transport>, TransportError> { /* S1/S3 */ }

```

| Subtask                                                                | id              | deliverable                     | acceptance                                                    | test_types |
|------------------------------------------------------------------------|-----------------|---------------------------------|---------------------------------------------------------------|------------|
| Trait +<br>TransportError<br>+<br>effective_mtu<br>contract            | HVPN-<br>P0-005 | lib.rs trait                    | cargo doc<br>clean; trait<br>object-safe                      | UT         |
| TransportConfig<br>enum + dial()<br>factory<br>skeleton                | HVPN-<br>P0-006 | dial()<br>matching<br>on config | returns<br>Box<dyn<br>Transport><br>for PlainUdp              | UT         |
| Transport<br>registry +<br>kind() label<br>table (for<br>metrics/logs) | HVPN-<br>P0-007 | registry<br>map<br>kind→ctor    | round-trips<br>label; no<br>per-flow<br>durable<br>state (I5) | UT, SEC    |

## TASK HVPN-P0-008 — Plain-UDP transport + loopback echo (the throughput baseline)

| Field  | Value               |
|--------|---------------------|
| parent | S0                  |
| kind   | task · type Feature |
| title  |                     |

| Field       | Value                                                                                                                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gate        | Implement plain-UDP Transport and prove a loopback echo round-trip                                                                                                                                                                                           |
| description | G1 (baseline that all transports are measured against)<br>Trivial PlainUdp over tokio::net::UdpSocket [04_P0 §4.3]; gives the <b>baseline</b> every other transport's ≥50% bar (G2) is computed against; loopback echo test validates the trait round-trips. |
| depends_on  | HVPN-P0-006                                                                                                                                                                                                                                                  |
| deliverable | helix-transport/src/plain_udp.rs + tests/loopback_echo.rs                                                                                                                                                                                                    |
| acceptance  | echo test sends 10k 1280-byte datagrams loopback, 0 lost, order-independent; captured CSV of round-trip count                                                                                                                                                |
| est_effort  | 0.5                                                                                                                                                                                                                                                          |
| test_types  | UT, IT, CONC, BENCH                                                                                                                                                                                                                                          |

```
// helix-transport/src/plain_udp.rs
pub struct PlainUdp { sock: tokio::net::UdpSocket, peer:
    SocketAddr }
#[async_trait]
impl Transport for PlainUdp {
    async fn send(&self, dg: Bytes) -> Result<(), TransportError>
    { self.sock.send_to(&dg, self.peer).await?; Ok(()) }
    async fn recv(&self) -> Result<Bytes, TransportError> {
        let mut buf = vec![0u8; 1500];
        let n = self.sock.recv(&mut buf).await?; buf.truncate(n);
        Ok(Bytes::from(buf))
    }
    fn kind(&self) -> &'static str { "plain-udp" }
    fn effective_mtu(&self) -> u16 { 1420 } // standard WG-over-IPv4
}
```

| Subtask                                                         | id              | deliverable                         | acceptance                                                        | test_types             |
|-----------------------------------------------------------------|-----------------|-------------------------------------|-------------------------------------------------------------------|------------------------|
| PlainUdp<br>send/recv<br>+<br>MTU=1420                          | HVPN-<br>P0-009 | plain_udp.rs                        | loopback<br>single<br>datagram<br>round-trips                     | UT, IT                 |
| Loopback<br>echo soak<br>(10k dg) +<br>datagram-<br>loss census | HVPN-<br>P0-010 | tests/<br>loopback_echo.rs<br>+ CSV | 0 lost @<br>loopback;<br>deterministic<br>over N=3<br>(\$11.4.50) | IT, CONC,<br>BENCH, FA |

## 4. Milestone S1 — helix-wg + helix-tun + orchestrator (HVPN-P0-011..021) · G1

**Proves:** WG core works through our transport — Linux client ↔ Linux gateway plain-UDP WG up; ping overlay IP. **Effort:** ~3d.

### TASK HVPN-P0-011 — helix-wg boringtun wrapper (handshake / encrypt / timers)

| Field       | Value                                                                                                                                                                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S1                                                                                                                                                                                                                                                         |
| kind        | task · type Feature                                                                                                                                                                                                                                        |
| title       | Wrap boringtun Tunn state machine into the helix-wg verdict API                                                                                                                                                                                            |
| gate        | G1                                                                                                                                                                                                                                                         |
| description | boringtun is the Phase-0 choice [04_P0 §4.4, research-boringtun]: pure-Rust userspace WG, no kernel module, identical on Linux/Android/iOS (the iOS path — no kernel WG on iOS). Wrap Tunn; pump its timer; route its four output verdicts into WgVerdict. |
| depends_on  | HVPN-P0-003                                                                                                                                                                                                                                                |
| deliverable | helix-wg/src/lib.rs (WgPeer + WgVerdict)                                                                                                                                                                                                                   |
| acceptance  | two WgPeers complete a Noise IK handshake in-process; data packet encrypts→decrypts to identical plaintext; captured test log                                                                                                                              |
| est_effort  | 1.5                                                                                                                                                                                                                                                        |
| test_types  | UT, CONC, RACE, MEM                                                                                                                                                                                                                                        |

```
// helix-wg/src/lib.rs (sketch) — survives into Phase 1 [04_P0 §4.4]
pub struct WgPeer { tunn:
    boringtun::noise::Tunn /* keys, endpoint */ }
pub enum WgVerdict { WriteToTransport(Bytes), WriteToTun(Bytes),
    Nothing, Err(WgError) }
impl WgPeer {
    pub fn handle_transport_in(&mut self, datagram: &[u8],
        scratch: &mut [u8]) -> WgVerdict { /* in: from transport */ }
    pub fn handle_tun_out(&mut self, ip_pkt: &[u8], scratch: &mut [u8]) -> WgVerdict { /* out: from TUN */ }
    pub fn tick(&mut self, scratch: &mut [u8]) -> WgVerdict { /* ~every 100-250ms: keepalive/rekey */ }
}
```

| Subtask | id | deliverable | acceptance | test_types |
|---------|----|-------------|------------|------------|
|         |    |             |            | UT, CONC   |

| Subtask                                                                    | id              | deliverable                  | acceptance                                                              | test_types |
|----------------------------------------------------------------------------|-----------------|------------------------------|-------------------------------------------------------------------------|------------|
| WgPeer::handle_transport_in /<br>handle_tun_out                            | HVPN-<br>P0-012 | encrypt/<br>decrypt<br>paths | in-process<br>handshake<br>+ data<br>round-trip                         | UT, RACE   |
| tick() timer pump<br>(keepalive/rekey)                                     | HVPN-<br>P0-013 | timer loop                   | rekey fires<br>after rekey-<br>after-time;<br>no deadlock<br>under loom |            |
| Scratch-buffer allocation<br>discipline (bounded, no per-<br>packet alloc) | HVPN-<br>P0-014 | reusable<br>scratch          | RSS flat<br>across 1M<br>packets (/proc<br>sample) —<br>feeds G3        | MEM, BENCH |

## TASK HVPN-P0-015 — helix-tun Linux TUN device abstraction

| Field       | Value                                                                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S1                                                                                                                                                                                                                                          |
| kind        | task · type Feature                                                                                                                                                                                                                         |
| title       | Provide the helix-tun Linux TUN device read/write abstraction                                                                                                                                                                               |
| gate        | G1                                                                                                                                                                                                                                          |
| description | OS TUN abstraction (Linux now; platform shims call in later via the same trait). Owns /dev/net/tun open, MTU/route config, async read/write of IP packets. On mobile the platform shim owns the TUN and feeds the same API [04_P0 §9 note]. |
| depends_on  | HVPN-P0-003                                                                                                                                                                                                                                 |
| deliverable | helix-tun/src/lib.rs + Linux impl                                                                                                                                                                                                           |
| acceptance  | brings up helix0 iface, assigns overlay IP, echoes an injected IP packet back to the reader; captured ip addr + pcap                                                                                                                        |
| est_effort  | 0.5                                                                                                                                                                                                                                         |
| test_types  | UT, IT                                                                                                                                                                                                                                      |

| Subtask                                                | id              | deliverable          | acceptance                                          | test_types |
|--------------------------------------------------------|-----------------|----------------------|-----------------------------------------------------|------------|
| TunDevice<br>trait +<br>Linux /dev/<br>net/tun<br>impl | HVPN-<br>P0-016 | lib.rs +<br>linux.rs | iface appears<br>in ip link;<br>read/write<br>works | IT         |

| Subtask                                   | id          | deliverable | acceptance                              | test_types |
|-------------------------------------------|-------------|-------------|-----------------------------------------|------------|
| MTU/route config + non-blocking async I/O | HVPN-P0-017 | config fns  | MTU=1420 set; no busy-spin (CONC check) | UT, CONC   |

## TASK HVPN-P0-018 — Orchestrator three-loop core + event stream + client/connector binaries

| Field       | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S1                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| kind        | task · type Feature                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| title       | Build the orchestrator three-loop core, event stream, and dual-mode binaries                                                                                                                                                                                                                                                                                                                                                                                           |
| gate        | G1                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| description | The orchestrator runs three loops — tun→wg→transport, transport→wg→tun, timer tick [04_P0 §4.5] — and exposes a tokio::sync::broadcast <b>event stream</b> (Connecting Handshaking Connected{transport,rtt} Reconnecting Down{reason}) that S5's FFI surfaces. -- mode={client connector} selects routing posture: client captures default route; connector advertises 10.10.0.0/24 and NAT-forwards decapsulated packets into veth-h. One binary per mode in Phase 0. |
| depends_on  | HVPN-P0-008, HVPN-P0-012, HVPN-P0-016                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| deliverable | helix-core/src/lib.rs + bin/helix-client.rs + bin/helix-connector.rs                                                                                                                                                                                                                                                                                                                                                                                                   |
| acceptance  | client orchestrator brings up plain-UDP WG to a second local orchestrator; ping <overlay-ip> succeeds; event stream emits Connected{plain-udp,rtt}; captured recording                                                                                                                                                                                                                                                                                                 |
| est_effort  | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| test_types  | UT, IT, E2E, CONC, RACE                                                                                                                                                                                                                                                                                                                                                                                                                                                |

The surviving status enum (mirrored across FFI in S5 [04\_P0 §4.5/§9]):

```
// helix-core: the event the whole UX reads from
pub enum TunnelStatus {
    Connecting, Handshaking,
    Connected { transport: String, rtt_ms: u32 },
    Reconnecting, Down { reason: String },
}
```

```
// client:    [app traffic] → TUN → WG encrypt → Transport →
              gateway
// connector: gateway → Transport → WG decrypt → forward into
              10.10.0.0/24 (and reverse)
```

| Subtask                                                            | id          | deliverable            | acceptance                                      | test_types     |
|--------------------------------------------------------------------|-------------|------------------------|-------------------------------------------------|----------------|
| Three async loops + broadcast event stream + TunnelStatus          | HVPN-P0-019 | orchestrator           | status transitions observed in test harness     | UT, CONC, RACE |
| helix-client binary (default-route capture, plain-UDP up)          | HVPN-P0-020 | bin/helix-client.rs    | client↔gateway WG up; ping overlay (G1 partial) | IT, E2E        |
| helix-connector binary (--mode=connector, advertise + NAT-forward) | HVPN-P0-021 | bin/helix-connector.rs | decapsulated packets reach veth-h and reverse   | IT, E2E        |

## 5. Milestone S2 — Connector advertise + two-way reach test (HVPN-P0-022..027) · G1

**Proves:** the two-way routing slice — connector advertises 10.10.0.0/24 (a netns LAN); client reaches 10.10.0.20:80. **Effort:** ~3d. **Closes G1.**

### TASK HVPN-P0-022 — netns + nftables + netem reproducible test rig

| Field       | Value                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S2                                                                                                                                                                                                                             |
| kind        | task · type Task                                                                                                                                                                                                               |
| title       | Stand up the reproducible netns LAN + DPI + impairment test rig                                                                                                                                                                |
| gate        | G1 (and G2 prep)                                                                                                                                                                                                               |
| description | Linux network namespaces simulate the connector's private LAN and a remote service so the whole slice runs on one box and is CI-scriptable later [04_P0 §3]. Three processes: client, gateway, connector. Includes the nft DPI |

| Field       | Value                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
|             | table (for G2) and tc netem impairment (for G2 loss tests), parked but installed here.                                                 |
| depends_on  | HVPN-P0-021                                                                                                                            |
| deliverable | rig/netns_up.sh, rig/netns_down.sh, rig/dpi_block.sh, rig/impair.sh                                                                    |
| acceptance  | rig/netns_up.sh creates lanA with a live http.server on 10.10.0.20:80; idempotent teardown; captured ip netns exec lanA curl localhost |
| est_effort  | 1.0                                                                                                                                    |
| test_types  | IT, FA, SEC                                                                                                                            |

The rig (concrete commands [04\_P0 \$3], to be wrapped in rig/\*.sh):

```
# rig/netns_up.sh – connector-side "private LAN" simulated with a netns
sudo ip netns add lanA
sudo ip link add veth-h type veth peer name veth-lanA
sudo ip link set veth-lanA netns lanA
sudo ip addr add 10.10.0.1/24 dev veth-h
# connector host side
sudo ip -n lanA addr add 10.10.0.20/24 dev veth-lanA # the "service" host
sudo ip link set veth-h up
sudo ip -n lanA link set veth-lanA up
sudo ip -n lanA route add default via 10.10.0.1
sudo ip netns exec lanA python3 -m http.server 80 --bind 10.10.0.20 & # the hello service

# rig/dpi_block.sh – DPI/censorship sim for G2: kill plain WG UDP, allow 443
sudo nft add table inet dpi
sudo nft add chain inet dpi fwd '{ type filter hook forward priority 0; }'
sudo nft add rule inet dpi fwd udp dport 51820 drop
sudo nft add rule inet dpi fwd udp dport 443 accept

# rig/impair.sh – impairment for loss-resilience (G2)
sudo tc qdisc add dev "$IFACE" root netem loss 5% delay 40ms 10ms
```

| Subtask                                   | id          | deliverable                   | acceptance                                                           | test_types |
|-------------------------------------------|-------------|-------------------------------|----------------------------------------------------------------------|------------|
| netns LAN + hello-service up/down scripts | HVPN-P0-023 | netns_up.sh/<br>netns_down.sh | LAN reachable from host; clean teardown leaves no orphan (\$11.4.14) | IT, FA     |

| Subtask                                                     | id          | deliverable                | acceptance                                                  | test_types |
|-------------------------------------------------------------|-------------|----------------------------|-------------------------------------------------------------|------------|
| nft DPI table + tc netem scripts (installed, parked for G2) | HVPN-P0-024 | dpi_block.sh/<br>impair.sh | nft list ruleset shows drop:51820/accept:443; netem applied | IT, SEC    |

## TASK HVPN-P0-025 — Two-way routing reach test (G1 close)

| Field       | Value                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S2                                                                                                                                                                                                                    |
| kind        | task · type Feature                                                                                                                                                                                                   |
| title       | Prove client reaches connector LAN host through the full plain-UDP chain                                                                                                                                              |
| gate        | G1                                                                                                                                                                                                                    |
| description | Success = curl from the client's overlay namespace to http://10.10.0.20/ returns the hello page, over plain-UDP [04_P0 §3]. This is the literal G1 close: real traffic client→gateway→connector LAN, both directions. |
| depends_on  | HVPN-P0-020, HVPN-P0-023                                                                                                                                                                                              |
| deliverable | rig/test_reach.sh + captured curl output + iperf3 CSV                                                                                                                                                                 |
| acceptance  | curl http://10.10.0.20/ returns hello page; iperf3 through-tunnel ≥ 80% bare link; deterministic over N=3 (§11.4.50); CSV + recording captured → gates.G1.outcome='pass'                                              |
| est_effort  | 1.0                                                                                                                                                                                                                   |
| test_types  | E2E, IT, FA, BENCH, SC, CH, HQA                                                                                                                                                                                       |

| Subtask                                          | id          | deliverable    | acceptance                                          | test_types        |
|--------------------------------------------------|-------------|----------------|-----------------------------------------------------|-------------------|
| Reach + return-path E2E (curl both directions)   | HVPN-P0-026 | test_reach.sh  | hello page returned; reverse path works             | E2E, IT           |
| iperf3 throughput vs bare-link baseline (≥80%) + | HVPN-P0-027 | iperf3 CSV row | ≥80% bar met; recovery <3s on iface flap (§11.4.85) | BENCH, SC, FA, CH |

| Subtask   | id | deliverable | acceptance | test_types |
|-----------|----|-------------|------------|------------|
| roam/flap |    |             |            |            |
| chaos     |    |             |            |            |

## 6. Milestone S3 — MASQUE / QUIC transport (HVPN-P0-028..038) · G2 ⚠ high-risk

**Proves:** the QUIC headline — same slice over :443/udp HTTP/3; works through a UDP-block rule. **Effort:** ~5d. **The riskiest Rust bet** [04\_P0 §5, §13].

**Byte flow** [04\_P0 §5.1, research-masque]:

WG datagram → HTTP Datagram (RFC 9297, ctx-id=0) → QUIC DATAGRAM frame (RFC 9221, unreliable) → QUIC/H3 to https://gateway:443 → edge extracts → WG datagram → kernel WG. The CONNECT-UDP request (RFC 9298) establishes the proxied UDP flow **once**; thereafter WG datagrams ride as HTTP Datagrams — no per-packet HTTP round trip. WG datagrams are independent & loss-tolerant = carry over QUIC **DATAGRAM frames**, never a QUIC stream (avoids head-of-line blocking) — getting this right is half the point of S3 (I2).

### TASK HVPN-P0-028 — quinn + h3 QUIC/HTTP-3 connection with unreliable datagrams

| Field       | Value                                                                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S3                                                                                                                                                                                                                                                                  |
| kind        | task · type Feature                                                                                                                                                                                                                                                 |
| title       | Establish a quinn QUIC/HTTP-3 connection exposing unreliable datagrams                                                                                                                                                                                              |
| gate        | G2                                                                                                                                                                                                                                                                  |
| description | quinn (mature QUIC, exposes send_datagram/read_datagram) + h3 (HTTP/3 on quinn, less battle-tested than Go's stack) [04_P0 §5.2, research-quinn]. Stand up the QUIC/H3 connection to gw:443 and verify unreliable datagram exchange before layering MASQUE framing. |
| depends_on  | HVPN-P0-006                                                                                                                                                                                                                                                         |
| deliverable | helix-transport/src/masque.rs (connection setup)                                                                                                                                                                                                                    |
| acceptance  | QUIC/H3 handshake to a local quinn server; send_datagram/read_datagram round-trips a 1200-byte payload; tshark shows QUIC long/short headers                                                                                                                        |

| Field      | Value       |
|------------|-------------|
| est_effort | 1.5         |
| test_types | UT, IT, SEC |

| Subtask                                              | id          | deliverable    | acceptance                                    | test_types |
|------------------------------------------------------|-------------|----------------|-----------------------------------------------|------------|
| quinn client+server endpoints + TLS/SNI config       | HVPN-P0-029 | endpoint setup | handshake completes; SNI = masquerade host    | IT, SEC    |
| Unreliable datagram round-trip (send_/read_datagram) | HVPN-P0-030 | datagram path  | 1200-byte payload round-trips; no stream used | UT, IT     |

## TASK HVPN-P0-031 — MASQUE CONNECT-UDP + HTTP-Datagram framing (the hand-rolled bet)

| Field       | Value                                                                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S3                                                                                                                                                                                                                                                                                                                                                     |
| kind        | task · type Feature                                                                                                                                                                                                                                                                                                                                    |
| title       | Hand-roll CONNECT-UDP and HTTP-Datagram framing over h3 + quinn datagrams                                                                                                                                                                                                                                                                              |
| gate        | G2, G4                                                                                                                                                                                                                                                                                                                                                 |
| description | In Rust, turnkey MASQUE/CONNECT-UDP is thin-to-absent — expect to implement the CONNECT-UDP request (RFC 9298) + HTTP-Datagram framing (RFC 9297) by hand on top of h3 + quinn datagrams [04_P0 §5.2]. <b>This relative immaturity vs Go's masque-go is itself a Phase-0 finding and a direct input to G4 (§7).</b> Record effort hours + bespoke LOC. |
| depends_on  | HVPN-P0-030                                                                                                                                                                                                                                                                                                                                            |
| deliverable | MasqueTransport impl + encode_http_datagram/decode_http_datagram                                                                                                                                                                                                                                                                                       |
| acceptance  | full slice works end-to-end over MASQUE (no DPI yet); effort/LOC recorded into the G4 decision input; captured pcap shows HTTP Datagram framing                                                                                                                                                                                                        |
| est_effort  | 2.0                                                                                                                                                                                                                                                                                                                                                    |
| test_types  | UT, IT, E2E, SEC                                                                                                                                                                                                                                                                                                                                       |

```
// helix-transport/src/masque.rs (the parts that matter) [04_P0
$5.2]
pub struct MasqueTransport { conn: quinn::Connection /* + CONNECT-
UDP flow ctx from dial() */ }
#[async_trait]
impl Transport for MasqueTransport {
    async fn send(&self, wg: Bytes) -> Result<(), TransportError>
    {
        let http_dg = encode_http_datagram(/*ctx*/0,
&wg); // RFC 9297 framing

        self.conn.send_datagram(http_dg)?; // RFC
9221 QUIC datagram
        Ok(())
    }
    async fn recv(&self) -> Result<Bytes, TransportError> {
        let dg = self.conn.read_datagram().await?;
        Ok(decode_http_datagram(dg)?) //
strip framing -> WG bytes
    }
    fn kind(&self) -> &'static str { "masque-h3" }
    fn effective_mtu(&self) -> u16 { 1280 } //
QUIC overhead eats headroom; measure & tune
}
}
```

| Subtask                                                               | id          | deliverable               | acceptance                                            | test_types |
|-----------------------------------------------------------------------|-------------|---------------------------|-------------------------------------------------------|------------|
| CONNECT-UDP request/response over h3 (RFC 9298)                       | HVPN-P0-032 | dial() Masque path        | proxied UDP flow established to gw internal WG socket | UT, IT     |
| HTTP-Datagram encode/decode (RFC 9297, ctx-id=0) + effective_mtu=1280 | HVPN-P0-033 | framing fns               | WG bytes survive encode→decode identical; MTU tuned   | UT, IT     |
| Bespoke-MASQUE effort + LOC record (G4 input)                         | HVPN-P0-034 | decision/masque_effort.md | hours + LOC captured; honest §11.4.6 finding          | CH, HQA    |

## TASK HVPN-P0-035 — DPI-block survival + loss resilience + wire-fingerprint (G2 close)

| Field  | Value                                                                 |
|--------|-----------------------------------------------------------------------|
| parent | S3                                                                    |
| kind   | task · type Feature                                                   |
| title  | Prove MASQUE survives a DPI UDP block, 5% loss, and looks like HTTP/3 |

| Field       | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gate        | G2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| description | S3 must demonstrate [04_P0 §5.3]: (1) slice works with plain WG UDP <b>blocked</b> (the nft rule) — real censorship evasion; (2) connection <b>looks like HTTP/3</b> to a passive observer (tshark: QUIC headers, no WG signature, SNI = masquerade host); (3) <b>loss resilience</b> — under netem loss 5%, MASQUE/QUIC sustains higher goodput than a UDP-over-TCP strawman; (4) <b>overhead quantified</b> vs plain-UDP for the auto-ladder cost model. Edge serves a believable decoy site to non-CONNECT-UDP probes (native masquerade, §5.4). |
| depends_on  | HVPN-P0-031, HVPN-P0-024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| deliverable | bench.sh MASQUE rows + tshark capture + decoy-page test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| acceptance  | with dpi_block.sh active, curl http://10.10.0.20/ still works over MASQUE; ≥50% of plain-UDP throughput; goodput@5%loss > UoT strawman; tshark classifies HTTP/3, no WG signature → gates.G2.outcome='pass'                                                                                                                                                                                                                                                                                                                                         |
| est_effort  | 1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| test_types  | E2E, SEC, BENCH, SC, LOAD, CH, HQA, FA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Subtask                                                       | id          | deliverable          | acceptance                                       | test_types      |
|---------------------------------------------------------------|-------------|----------------------|--------------------------------------------------|-----------------|
| End-to-end through DPI UDP block (51820 dropped, 443 allowed) | HVPN-P0-036 | reach test under DPI | hello page returned MASQUE-only                  | E2E, SEC, FA    |
| Loss-resilience vs UoT strawman + overhead quantification     | HVPN-P0-037 | bench.sh rows        | MASQUE goodput@5% > UoT; overhead CSV captured   | BENCH, SC, LOAD |
| Wire-fingerprint (tshark) + native decoy site for probes      | HVPN-P0-038 | pcap + decoy page    | classified HTTP/3, no WG sig; probe → decoy page | SEC, CH, HQA    |

## 7. Milestone S4 — Go vs Rust gateway edge + A/B benchmark (HVPN-P0-039..048) · G4 ⚠

**Proves:** the edge decision — build **both** edges for MASQUE termination, decide on numbers [04\_P0 §7]. **Effort:** ~4d. Identical job: terminate MASQUE on :443, hand WG to kernel WG.

|                 | Rust edge (helix-edge)                                   | Go edge                                             |
|-----------------|----------------------------------------------------------|-----------------------------------------------------|
| QUIC/H3         | quinn + h3 (+ hand-rolled CONNECT-UDP)                   | quic-go + <b>masque-go</b> (turnkey CONNECT-UDP/IP) |
| Code reuse      | <b>Shares helix-transport byte-for-byte with clients</b> | separate impl; must track the Rust one              |
| Fits stack      | new language in the server tree                          | matches the Go/Gin control plane exactly            |
| Hot-path cost   | no GC                                                    | GC pauses under load (measure)                      |
| MASQUE maturity | thinner ecosystem (a Phase-0 risk)                       | more mature MASQUE tooling                          |

### TASK HVPN-P0-039 — Rust edge helix-edge (MASQUE termination, shares helix-transport)

| Field       | Value                                                                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S4                                                                                                                                                                                                        |
| kind        | task · type Feature                                                                                                                                                                                       |
| title       | Build the Rust helix-edge terminating MASQUE and handing WG to kernel                                                                                                                                     |
| gate        | G4                                                                                                                                                                                                        |
| description | Rust contender [04_P0 §7.1]: reuses helix-transport byte-for-byte — the single-implementation guarantee; terminates MASQUE on :443, hands extracted WG datagrams to kernel WG / boringtun gateway socket. |
| depends_on  | HVPN-P0-031                                                                                                                                                                                               |
| deliverable | helix-edge/ Rust binary                                                                                                                                                                                   |
| acceptance  | terminates the S3 client's MASQUE flow; slice reaches 10.10.0.20:80; captured reach test                                                                                                                  |
| est_effort  | 1.0                                                                                                                                                                                                       |
| test_types  | IT, E2E, SEC, RACE                                                                                                                                                                                        |

| Subtask                                     | id          | deliverable   | acceptance                           | test_types |
|---------------------------------------------|-------------|---------------|--------------------------------------|------------|
| MASQUE listener + WG handoff to kernel WG   | HVPN-P0-040 | edge main     | slice reaches LAN host via Rust edge | IT, E2E    |
| Native decoy masquerade for non-flow probes | HVPN-P0-041 | decoy handler | probe → website; valid flow → tunnel | SEC        |

## TASK HVPN-P0-042 — Go edge (quic-go + masque-go, turnkey CONNECT-UDP)

| Field       | Value                                                                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S4                                                                                                                                                         |
| kind        | task · type Feature                                                                                                                                        |
| title       | Build the Go edge with quic-go and masque-go for MASQUE termination                                                                                        |
| gate        | G4                                                                                                                                                         |
| description | Go contender [04_P0 §7.1]: quic-go + masque-go turnkey CONNECT-UDP/IP; matches the Go/Gin control plane exactly. Same job as the Rust edge for a fair A/B. |
| depends_on  | HVPN-P0-024                                                                                                                                                |
| deliverable | go-edge/ Go binary                                                                                                                                         |
| acceptance  | terminates the same MASQUE flow; slice reaches 10.10.0.20:80; -race clean                                                                                  |
| est_effort  | 1.0                                                                                                                                                        |
| test_types  | IT, E2E, SEC, RACE                                                                                                                                         |

| Subtask                                          | id          | deliverable           | acceptance                                         | test_types |
|--------------------------------------------------|-------------|-----------------------|----------------------------------------------------|------------|
| quic-go + masque-go termination + WG handoff     | HVPN-P0-043 | go edge main          | slice reaches LAN host via Go edge                 | IT, E2E    |
| go test -race + dev-cost hours record (G4 input) | HVPN-P0-044 | decision/go_effort.md | -race clean; hours-to-correct-CONNECT-UDP captured | RACE, CH   |

## TASK HVPN-P0-045 — bench.sh A/B harness + decision matrix (G4 close)

| Field       | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| kind        | task · type Feature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| title       | Run the identical A/B benchmark and record the edge-language decision                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| gate        | G4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| description | One harness, both edges, same rig, same kernel-WG backend [04_P0 §7.2]: throughput (1/10/100 clients), CPU per Gbps (cost-to-serve), p50/p99 RTT (Go GC tail), connection churn (handshakes/sec), memory under churn, dev cost. Fill the decision matrix; record the call in the decision log (§12). <b>Default lean:</b> Rust if within ~10–15% on throughput/CPU and hand-rolled CONNECT-UDP isn't a quagmire (single-impl win); else Go + accept dual MASQUE impls (mitigated by shared test vectors + conformance suite). |
| depends_on  | HVPN-P0-040, HVPN-P0-043                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| deliverable | bench.sh + results/edge_ab.csv + filled decision matrix                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| acceptance  | CSV with CPU/Gbps, p99, churn, memory for both edges over N=3 (§11.4.50); decision recorded → gates.G4.outcome IN ('rust', 'go') with evidence path                                                                                                                                                                                                                                                                                                                                                                           |
| est_effort  | 1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| test_types  | BENCH, LOAD, SC, FA, CH, HQA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

bench.sh outline (one script drives S4 A/B **and** S7 transport pair → CSV [04\_P0 §8]):

```
#!/usr/bin/env bash
# bench.sh – comparable results for every transport × edge
#               combination. Outputs one CSV row/run.
set -euo pipefail
EDGE="$1"          # rust-edge | go-edge
TRANSPORT="$2"     # plain-udp | masque-h3
CLIENTS="${3:-1}"  # 1 | 10 | 100
RUN="${4:-1}"      # iteration for §11.4.50 determinism (run N=3)
OUT="results/bench.csv"

bring_up_rig       # rig/netns_up.sh (lanA + hello service)
start_edge "$EDGE" # exclusive owner of :443 (§11.4.119)
start_client "$TRANSPORT" --clients "$CLIENTS"
```

```

THRPUT=$(iperf3 -J -c 10.10.0.20 -P "$CLIENTS" | jq
    '.end.sum_received.bits_per_second')
ADDED_RTT_P50=$(ping_overlay_p50); ADDED_RTT_P99=$(
    ping_overlay_p99)
GOODPUT_LOSS=$(with_netem 'loss 5% delay 40ms' iperf3_goodput)
    # G2 / SC
HS_TIME=$(timestamp_connect_to_first_data)
RECONNECT=$(flap_iface_time_to_recovery)
    # SC, <3s
CPU_PER_GBPS=$(pidstat_at_saturation "$EDGE")
    # cost-to-serve
EDGE_RSS=$(proc_rss_of "$EDGE")
CORE_RSS=$(proc_rss_of helix-client)
    # /proc; iOS via Instruments (§10)
FINGERPRINT=$(tshark_classify_flow)
    # http3 | wg-signature (SEC)

printf '%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s\n' \
    "$RUN" "$EDGE" "$TRANSPORT" "$CLIENTS" "$THRPUT"
    "$ADDED_RTT_P50" "$ADDED_RTT_P99" \
    "$GOODPUT_LOSS" "$HS_TIME" "$RECONNECT" "$CPU_PER_GBPS"
    "$EDGE_RSS" "$FINGERPRINT" >> "$OUT"
tear_down_rig          # rig/netns_down.sh – leave quiescent
    (§11.4.14)

```

| Subtask                                                                            | id              | deliverable        | acceptance                                                           | test_types         |
|------------------------------------------------------------------------------------|-----------------|--------------------|----------------------------------------------------------------------|--------------------|
| bench.sh<br>harness +<br>CSV schema<br>(transport ×<br>edge ×<br>clients ×<br>run) | HVPN-<br>P0-046 | bench.sh           | runs all<br>combos;<br>deterministic<br>CSV over<br>N=3              | BENCH, FA          |
| Throughput/<br>CPU-per-<br>Gbps/p99/<br>churn/<br>memory<br>matrix fill            | HVPN-<br>P0-047 | edge_ab.csv        | all six<br>metrics<br>captured<br>both edges;<br>churn under<br>LOAD | BENCH, LOAD,<br>SC |
| Decision-<br>matrix<br>scoring +<br>decision-log<br>entry (§12)                    | HVPN-<br>P0-048 | decision/<br>G4.md | weighted<br>matrix →<br>recorded call<br>+<br>consequence            | CH, HQA            |

## 8. Milestone S5 — flutter\_rust\_bridge FFI + Flutter-Linux toggle (HVPN-P0-049..054) · G5

**Proves:** the FFI boundary is clean enough to drive the core from Dart. **Effort:** ~3d. Use **flutter\_rust\_bridge v2** — generates Dart $\rightleftharpoons$ Rust glue, handles async + streams; the surface stays tiny and stable [04\_P0 §9].

### TASK HVPN-P0-049 — helix-ffi flutter\_rust\_bridge v2 surface (survivor FFI)

| Field       | Value                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S5                                                                                                                                                                                                                                           |
| kind        | task · type Feature                                                                                                                                                                                                                          |
| title       | Define the helix-ffi flutter_rust_bridge surface that survives into Phase 1                                                                                                                                                                  |
| gate        | G5                                                                                                                                                                                                                                           |
| description | The FFI surface graduates into Phase 1 (gains policy/multi-hop fields) [04_P0 §14]. start(cfg) spins the orchestrator; stop() tears down; status_stream(sink) forwards the §4.5 broadcast channel; TunnelStatus mirrored via #[frb(mirror)]. |
| depends_on  | HVPN-P0-019                                                                                                                                                                                                                                  |
| deliverable | helix-ffi/src/api.rs + generated Dart bindings                                                                                                                                                                                               |
| acceptance  | flutter_rust_bridge_codegen produces Dart bindings that compile; start/stop/status_stream callable from a Dart unit test                                                                                                                     |
| est_effort  | 1.5                                                                                                                                                                                                                                          |
| test_types  | UT, IT                                                                                                                                                                                                                                       |

```
// helix-ffi/src/api.rs – flutter_rust_bridge v2 generates Dart
// bindings [04_P0 §9]
pub struct ClientConfig { pub map_path: String, pub transport:
    String /* auto|plain|masque */ }
pub async fn start(cfg: ClientConfig) -> anyhow::Result<()> { /*
    spin up helix-core orchestrator */ }
pub async fn stop() -> anyhow::Result<()> { /* tear down */ }
/// Live status as a stream the UI subscribes to (maps to the §4.5
    broadcast channel).
pub fn status_stream(sink: StreamSink<TunnelStatus>) { /* forward
    core events */ }
#[frb(mirror)]
pub enum TunnelStatus {
    Connecting, Handshaking,
    Connected { transport: String, rtt_ms: u32 },
```

```
Reconnecting, Down { reason: String },
}
```

| Subtask                                              | id              | deliverable          | acceptance                                           | test_types |
|------------------------------------------------------|-----------------|----------------------|------------------------------------------------------|------------|
| start/stop/<br>ClientConfig +<br>codegen             | HVPN-<br>P0-050 | api.rs +<br>bindings | Dart can<br>call start/<br>stop;<br>codegen<br>clean | UT, IT     |
| status_stream<br>+<br>#[frb(mirror)]<br>TunnelStatus | HVPN-<br>P0-051 | stream<br>binding    | Dart<br>receives<br>mapped<br>status<br>events       | UT, IT     |

## TASK HVPN-P0-052 — Flutter-Linux toggle + live status chip (G5 close)

| Field       | Value                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S5                                                                                                                                                                                                                                                                                                                        |
| kind        | task · type Feature                                                                                                                                                                                                                                                                                                       |
| title       | Ship a Flutter-Linux window with a connect toggle and live status chip                                                                                                                                                                                                                                                    |
| gate        | G5                                                                                                                                                                                                                                                                                                                        |
| description | G5 pass [04_P0 §9]: a Flutter-Linux window with one connect/disconnect toggle and a status chip going Connecting → Handshaking → Connected (masque, 23ms) driven <b>only</b> by the Rust event stream. On Linux the core drives the TUN directly; on mobile the platform shim owns the TUN and the same FFI drives logic. |
| depends_on  | HVPN-P0-051, HVPN-P0-025                                                                                                                                                                                                                                                                                                  |
| deliverable | app_access/ minimal Flutter-Linux app                                                                                                                                                                                                                                                                                     |
| acceptance  | window toggle brings tunnel up/down; chip updates live from the core stream; window-scoped recording (§11.4.159) showing the transition → gates.G5.outcome='pass'                                                                                                                                                         |
| est_effort  | 1.0                                                                                                                                                                                                                                                                                                                       |
| test_types  | UT, E2E, FA, CH, HQA                                                                                                                                                                                                                                                                                                      |

```
// app_access – the entire happy path the spike must show [04_P0
  §9]
await HelixCore.start(ClientConfig(mapPath: '/etc/helix/
  map.json', transport: 'auto'));
HelixCore.statusStream().listen((s) => setState(() => status =
  s)); // live chip updates
```

```
// ...toggle off:
await HelixCore.stop();
```

| Subtask                                                  | id          | deliverable       | acceptance                                                           | test_types       |
|----------------------------------------------------------|-------------|-------------------|----------------------------------------------------------------------|------------------|
| Toggle button → start/stop wiring                        | HVPN-P0-053 | toggle widget     | tunnel up/down from UI                                               | E2E              |
| StatusChip bound to statusStream + window-scoped capture | HVPN-P0-054 | chip widget + mp4 | chip shows Connected(masque, NNms); recording validated (\$11.4.163) | E2E, FA, CH, HQA |

## 9. Milestone S6 — Android VpnService + JNI (HVPN-P0-055..060)

**Proves:** the mobile path is real — Android VpnService + JNI loads helix-core; tunnel up on a real phone. **Effort:** ~3d. (Prerequisite for S7's device work.)

### TASK HVPN-P0-055 — Android VpnService + JNI bridge to helix-core

| Field       | Value                                                                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S6                                                                                                                                                                                                           |
| kind        | task · type Feature                                                                                                                                                                                          |
| title       | Build the Android VpnService and JNI bridge loading the Rust helix-core                                                                                                                                      |
| gate        | — (mobile path; feeds G3)                                                                                                                                                                                    |
| description | Android VpnService owns the TUN fd; JNI loads helix-core .so (aarch64-linux-android); the platform shim feeds packets to the same orchestrator API [04_P0 §2 S6, §9 note]. UniFFI generates the Kotlin shim. |
| depends_on  | HVPN-P0-019                                                                                                                                                                                                  |
| deliverable | shims/android/ app + libhelix_core.so                                                                                                                                                                        |
| acceptance  | app builds; VpnService establishes the TUN; JNI calls reach the core; captured adb logcat                                                                                                                    |
| est_effort  | 1.5                                                                                                                                                                                                          |
| test_types  | UT, IT, MEM                                                                                                                                                                                                  |

| Subtask                                                  | id          | deliverable     | acceptance                                          | test_types |
|----------------------------------------------------------|-------------|-----------------|-----------------------------------------------------|------------|
| aarch64-linux-android staticlib + UniFFI Kotlin bindings | HVPN-P0-056 | .so + bindings  | JNI call into core returns; builds in CI-less local | UT, IT     |
| VpnService TUN ownership + packet pump to core           | HVPN-P0-057 | VpnService impl | TUN fd handed to core; packets flow                 | IT, MEM    |

## TASK HVPN-P0-058 — Tunnel up on a real Android phone (E2E reach)

| Field       | Value                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S6                                                                                                                                                                                                   |
| kind        | task · type Feature                                                                                                                                                                                  |
| title       | Bring the tunnel up on a physical Android device and reach the LAN host                                                                                                                              |
| gate        | — (feeds G3 transport memory baselines)                                                                                                                                                              |
| description | Real device, not emulator (memory/behaviour representative). Tunnel up, reach 10.10.0.20:80 via plain-UDP then MASQUE; dumpsys meminfo RSS recorded as an Android comparison point for the iOS gate. |
| depends_on  | HVPN-P0-057, HVPN-P0-035                                                                                                                                                                             |
| deliverable | device recording + dumpsys meminfo capture                                                                                                                                                           |
| acceptance  | curl/browser on phone reaches LAN host both transports; RSS captured; §11.4.144 device-availability followed during the run                                                                          |
| est_effort  | 1.5                                                                                                                                                                                                  |
| test_types  | E2E, MEM, SC, CH, HQA                                                                                                                                                                                |

| Subtask                            | id          | deliverable      | acceptance                         | test_types   |
|------------------------------------|-------------|------------------|------------------------------------|--------------|
| Plain-UDP + MASQUE reach on device | HVPN-P0-059 | device recording | LAN host reachable both transports | E2E, CH      |
| dumpsys meminfo                    | HVPN-P0-060 | meminfo CSV      | RSS captured; survives             | MEM, SC, HQA |

| Subtask                         | id | deliverable | acceptance            | test_types |
|---------------------------------|----|-------------|-----------------------|------------|
| RSS<br>sampling +<br>roam chaos |    |             | WiFi↔LTE<br>roam (SC) |            |

## 10. Milestone S7 — iOS

### NEPacketTunnelProvider + memory harness (HVPN-P0-061..070) · G3 ⚠ make-or-break

**Proves:** the make-or-break gate — Rust core lives inside a NEPacketTunnelProvider under its memory budget with headroom.

**Effort:** ~4d. If the core can't live in the NE, the "Rust core on every platform" thesis is wounded [04\_P0 §6].

**The constraint** [04\_P0 §6.1]: a Network Extension's packet-tunnel process runs with a **far tighter memory budget than the host app** — historically ~15 MB, raised on newer iOS but still strict and **silently fatal**: exceed it and iOS kills the extension, dropping the tunnel. The exact ceiling is device/OS-dependent and **MUST be measured, not assumed** (§11.4.6). This tight budget is the single strongest reason the data plane is **Rust** (no GC, bounded allocations) rather than Go (D2 [SYNTHESIS]).

#### Harness layout [04\_P0 §6.2]

```
ios-spike/
├── HelixTunnel/                                # NEPacketTunnelProvider (Swift)
│   └── PacketTunnelProvider.swift
├── libhelix_core.a                             # Rust staticlib, aarch64-apple-
ios, --release + LTO + strip
└── helix_core.h                               # cbindgen header (or UniFFI-
generated)
```

*// PacketTunnelProvider.swift (skeleton) [04\_P0 §6.2]*

```
import NetworkExtension
class PacketTunnelProvider: NEPacketTunnelProvider {
    override func startTunnel(options: [String:NSObject]?,
        completionHandler: @escaping (Error?)->Void) {
        // 1. configure NEPacketTunnelNetworkSettings (overlay IP,
        //     routes, DNS)
        // 2. helix_core_start(config)           // Rust core takes over
        //     packetFlow read/write
        // 3. pump: packetFlow.readPackets -> helix_core_tun_out;
        //     core -> packetFlow.writePackets
```

```

    }
}

```

## TASK HVPN-P0-061 — iOS staticlib build (size-optimised) + cbindgen/UniFFI header

| Field       | Value                                                                                                                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S7                                                                                                                                                                                                                                  |
| kind        | task · type Feature                                                                                                                                                                                                                 |
| title       | Build the size-optimised aarch64-apple-ios staticlib and FFI header                                                                                                                                                                 |
| gate        | G3                                                                                                                                                                                                                                  |
| description | Build the <b>same helix-core</b> as aarch64-apple-ios staticlib: opt-level=z/s + LTO + panic=abort + strip; record binary size [04_P0 §6.3]. cbindgen (or UniFFI) generates helix_core.h. Binary size is itself a recorded finding. |
| depends_on  | HVPN-P0-014, HVPN-P0-019                                                                                                                                                                                                            |
| deliverable | libhelix_core.a + helix_core.h + size report                                                                                                                                                                                        |
| acceptance  | staticlib builds for aarch64-apple-ios; size recorded; symbols stripped; captured size output                                                                                                                                       |
| est_effort  | 1.0                                                                                                                                                                                                                                 |
| test_types  | UT, MEM, BENCH                                                                                                                                                                                                                      |

| Subtask                                                                      | id          | deliverable           | acceptance                                            | test_types |
|------------------------------------------------------------------------------|-------------|-----------------------|-------------------------------------------------------|------------|
| Release profile<br>(opt-level=z,<br>LTO,<br>panic=abort,<br>strip)           | HVPN-P0-062 | Cargo.toml<br>profile | builds; size<br>captured                              | UT, BENCH  |
| cbindgen/UniFFI<br>header<br>helix_core.h<br>(helix_core_start/<br>_tun_out) | HVPN-P0-063 | header                | header<br>compiles in<br>Xcode;<br>symbols<br>resolve | UT, MEM    |

## TASK HVPN-P0-064 — NEPacketTunnelProvider skeleton + packetFlow pump

| Field  | Value                                                                 |
|--------|-----------------------------------------------------------------------|
| parent | S7                                                                    |
| kind   | task · type Feature                                                   |
| title  | Wire the NEPacketTunnelProvider to drive the Rust core via packetFlow |

| Field       | Value                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gate        | G3                                                                                                                                                                                                                                                     |
| description | Configure NEPacketTunnelNetworkSettings (overlay IP, routes, DNS); helix_core_start(config); pump packetFlow.readPackets → helix_core_tun_out and core → packetFlow.writePackets [04_P0 §6.2]. The shim owns the TUN; the Rust core owns the datapath. |
| depends_on  | HVPN-P0-063                                                                                                                                                                                                                                            |
| deliverable | ios-spike/HelixTunnel/PacketTunnelProvider.swift                                                                                                                                                                                                       |
| acceptance  | tunnel establishes on a real device; curl/Safari reaches 10.10.0.20:80 over plain-UDP; captured device recording                                                                                                                                       |
| est_effort  | 1.0                                                                                                                                                                                                                                                    |
| test_types  | IT, E2E, MEM                                                                                                                                                                                                                                           |

| Subtask                                            | id          | deliverable | acceptance                              | test_types |
|----------------------------------------------------|-------------|-------------|-----------------------------------------|------------|
| NEPacketTunnelNetworkSettings config + startTunnel | HVPN-P0-065 | provider    | tunnel up; routes/DNS applied           | IT, E2E    |
| packetFlow ⇌ core pump (read/write)                | HVPN-P0-066 | pump loop   | LAN host reachable on device, plain-UDP | E2E, MEM   |

## TASK HVPN-P0-067 — Memory harness: 1 GB / 30-min soak, both transports (G3 verdict)

| Field       | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| kind        | task · type Feature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| title       | Run the rigorous iOS memory soak deciding the Rust-on-every-platform thesis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| gate        | G3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| description | Be rigorous — this number decides architecture [04_P0 §6.3]. Drive a <b>sustained 1 GB transfer</b> through the tunnel on a <b>real device</b> (Simulator memory is not representative). Sample the <b>extension process</b> RSS via Xcode Instruments (Allocations + VM Tracker), not the host app — peak and steady-state. Run <b>plain-UDP</b> and <b>MASQUE</b> separately (QUIC buffers cost more memory; both must pass). <b>Pass:</b> steady-state peak under the device-enforced ceiling with <b>≥30% headroom</b> for both transports across a 30-minute |

| Field       | Value                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | soak. <b>If it fails</b> [04_P0 §6.4], record fallback ladder: (1) shrink buffers / cap QUIC flow-control windows in the iOS build; (2) move MASQUE off-device for iOS (plain WG + on-path obfs only — partial feature loss); (3) split the core so only the lean WG datapath is in-extension, QUIC negotiation in the app via app-extension IPC. Each fallback is a real product decision. |
| depends_on  | HVPN-P0-066, HVPN-P0-035                                                                                                                                                                                                                                                                                                                                                                    |
| deliverable | ios-spike/memory_report.md (peak/steady RSS, both transports, 30-min) + Instruments trace                                                                                                                                                                                                                                                                                                   |
| acceptance  | RSS report shows steady-state peak vs ceiling + headroom % for plain-UDP <b>and</b> MASQUE; verdict recorded → gates.G3.outcome IN ('pass', 'fail'); if fail, the §6.4 ladder choice is recorded (§11.4.112-style honest finding)                                                                                                                                                           |
| est_effort  | 2.0                                                                                                                                                                                                                                                                                                                                                                                         |
| test_types  | MEM, SC, BENCH, FA, CH, HQA                                                                                                                                                                                                                                                                                                                                                                 |

The honest-procedure checklist (each is a falsifiable evidence row, §11.4.5/.107):

1. Build same-core staticlib (size-optimised) — done in HVPN-P0-062.
2. Real device, **not** Simulator — recorded device serial (§11.4.144 followed).
3. Sustained 1 GB transfer through tunnel — iperf3/bulk HTTP via the rig.
4. Instruments on the **extension** process (Allocations + VM Tracker) — peak + steady RSS.
5. Plain-UDP run AND MASQUE run, 30-min soak each.
6. Pass = steady peak < ceiling with  $\geq 30\%$  headroom, **both** transports.
7. On fail → record §6.4 fallback ladder choice as the architectural consequence.

| Subtask                                                  | id              | deliverable        | acceptance                                          | test_types        |
|----------------------------------------------------------|-----------------|--------------------|-----------------------------------------------------|-------------------|
| Plain-UDP<br>30-min/1<br>GB soak +<br>Instruments<br>RSS | HVPN-<br>P0-068 | trace + RSS<br>row | steady peak<br>+ headroom<br>% captured<br>(plain)  | MEM, SC,<br>BENCH |
| MASQUE<br>30-min/1<br>GB soak +<br>Instruments<br>RSS    | HVPN-<br>P0-069 | trace + RSS<br>row | steady peak<br>+ headroom<br>% captured<br>(masque) | MEM, SC,<br>BENCH |

| Subtask                                             | id          | deliverable      | acceptance                                             | test_types  |
|-----------------------------------------------------|-------------|------------------|--------------------------------------------------------|-------------|
| Verdict + §6.4 fallback-ladder decision record (G3) | HVPN-P0-070 | memory_report.md | go/no-go recorded with headroom; ladder choice if fail | CH, HQA, FA |

## 11. Milestone S8 — Static network-map ingest + reconcile (HVPN-P0-071..076) • G6

**Proves:** the real-time model — fake the Go control plane with a file of the **exact shape** Phase 1's WatchNetworkMap will stream, so the reconciler is real [04\_P0 §10]. **Effort:** ~2d. The literal seed of the Phase-1 push model.

```
// /etc/helix/map.json — the desired-state document the core
// reconciles to [04_P0 §10]
{
  "self": { "overlay_ip": "fd7a:helix:1::2/128", "transport":
    "auto" },
  "gateway": { "endpoint": "gw.example:443", "wg_pubkey": "...",
    "masque_sni": "cdn.example" },
  "peers": [ { "name": "connectorA", "wg_pubkey": "...",
    "allowed_ips": ["10.10.0.0/24"] } ],
  "dns": ["fd7a:helix:1::1"]
}
```

### TASK HVPN-P0-071 — map.json schema + reconciler diff engine

| Field       | Value                                                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S8                                                                                                                                                                                                                                                                                                            |
| kind        | task · type Feature                                                                                                                                                                                                                                                                                           |
| title       | Define the map.json desired-state schema and the reconciler diff engine                                                                                                                                                                                                                                       |
| gate        | G6                                                                                                                                                                                                                                                                                                            |
| description | The file has the exact shape Phase 1's WatchNetworkMap protobuf streams [04_P0 §10, §14] — so the reconciler graduates unchanged. The reconciler diffs desired-vs-actual and converges (add/remove peers, switch transport) <b>without restarting</b> the process; a file-watch stands in for a stream event. |

| Field       | Value                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------|
| depends_on  | HVPN-P0-019                                                                                                     |
| deliverable | helix-core/src/map.rs (schema + reconcile diff)                                                                 |
| acceptance  | reconciler computes minimal add/remove deltas from two map versions; unit-tested; no restart of unrelated state |
| est_effort  | 1.0                                                                                                             |
| test_types  | UT, CONC, RACE                                                                                                  |

| Subtask                                                                                                                                                       | id              | deliverable  | acceptance                                                       | test_types        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------|------------------------------------------------------------------|-------------------|
| map.json<br>serde<br>schema<br>(self/<br>gateway/<br>peers/<br>dns)<br>Reconcile<br>diff<br>(desired<br>vs actual<br>peers/<br>transport)<br>+ file-<br>watch | HVPN-<br>P0-072 | map.rs types | parses the \$10<br>document; shape<br>matches<br>WatchNetworkMap | UT                |
|                                                                                                                                                               | HVPN-<br>P0-073 | reconcile fn | minimal deltas;<br>idempotent; no full<br>teardown               | UT, CONC,<br>RACE |

## TASK HVPN-P0-074 — Map-delta reconcile, no-restart, new network reachable (G6 close)

| Field       | Value                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | S8                                                                                                                                                                                                                                 |
| kind        | task · type Feature                                                                                                                                                                                                                |
| title       | Prove a map delta brings a new network reachable without restart                                                                                                                                                                   |
| gate        | G6                                                                                                                                                                                                                                 |
| description | G6 pass [04_P0 \$10]: edit map.json to add the connector's prefix → the core picks up the delta (file-watch standing in for a stream event) and the client can now reach 10.10.0.20 — no restart, no reconnect of unrelated state. |
| depends_on  | HVPN-P0-073, HVPN-P0-025                                                                                                                                                                                                           |
| deliverable | rig/test_reconcile.sh + before/after reach recording                                                                                                                                                                               |
| acceptance  |                                                                                                                                                                                                                                    |

| Field      | Value                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
|            | before: curl 10.10.0.20 fails (no peer); after map edit: succeeds, same process PID, tunnel never dropped → gates.G6.outcome='pass' |
| est_effort | 1.0                                                                                                                                 |
| test_types | E2E, IT, FA, SC, CH, HQA                                                                                                            |

| Subtask                                          | id          | deliverable    | acceptance                                  | test_types  |
|--------------------------------------------------|-------------|----------------|---------------------------------------------|-------------|
| Add-peer delta → reachable, PID unchanged        | HVPN-P0-075 | reconcile test | new prefix reachable; same PID; no flap     | E2E, IT, FA |
| Remove-peer delta + chaos (corrupt map mid-edit) | HVPN-P0-076 | chaos test     | bad map rejected, last-good kept (§11.4.85) | SC, CH, HQA |

## 12. Cross-cutting deliverables (HVPN-P0-077..080)

These span all milestones — the harness, the decision log, the DB/CI wiring, and the Helix-ecosystem infra. They are Phase-0 definition-of-done items [04\_P0 §11].

### TASK HVPN-P0-077 — Unified measurement harness + comparable CSV methodology

| Field       | Value                                                                                                                                                                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | —                                                                                                                                                                                                                                                                                                                                      |
| kind        | task · type Task                                                                                                                                                                                                                                                                                                                       |
| title       | Provide the unified measurement methodology and comparable CSV outputs                                                                                                                                                                                                                                                                 |
| gate        | G1, G2, G4                                                                                                                                                                                                                                                                                                                             |
| description | One harness, run for every transport × edge combination so results are comparable [04_P0 §8]: throughput, added latency (p50/p99), loss resilience, handshake time, reconnect/roam, edge CPU/Gbps, core RSS per platform, wire fingerprint. bench.sh (HVPN-P0-046) is the engine; this item owns the <b>methodology + pass bars</b> as |

| Field       | Value                                                                                   |
|-------------|-----------------------------------------------------------------------------------------|
|             | the single comparable contract so the decision tables fill themselves.                  |
| depends_on  | HVPN-P0-046                                                                             |
| deliverable | docs/spike/measurement.md (the §8 metric table as a contract)                           |
| acceptance  | every gate's numbers trace to a methodology row; pass-bars match §1; CSV columns stable |
| est_effort  | 0.5                                                                                     |
| test_types  | BENCH, FA, CH                                                                           |

Measurement contract [04\_P0 §8] (the pass bars the CSV is judged against):

| Metric                    | How                                                  | Pass bar                                                         |
|---------------------------|------------------------------------------------------|------------------------------------------------------------------|
| Through-tunnel throughput | iperf3 -c <lan-host-via-tunnel> (TCP+UDP)            | plain-UDP $\geq 80\%$ bare link; MASQUE $\geq 50\%$ of plain-UDP |
| Added latency             | ping overlay vs bare; p50/p99                        | plain-UDP <2 ms; MASQUE <15 ms added                             |
| Loss resilience           | tc netem loss 5% delay 40ms; goodput                 | MASQUE/QUIC > UoT strawman                                       |
| Handshake time            | connect→first-data                                   | <1 s plain; <2 s MASQUE                                          |
| Reconnect/roam            | flap client iface; time to recovery                  | <3 s                                                             |
| Edge CPU per Gbps         | pidstat/perf at saturation                           | record (cost-to-serve)                                           |
| Core RSS per platform     | /proc (Linux), Instruments (iOS), dumphsys (Android) | iOS gate per §10                                                 |
| Wire fingerprint          | tshark of MASQUE flow                                | classified HTTP/3, no WG signature                               |

## TASK HVPN-P0-078 — Decision log + 5-minute demo script

| Field  | Value                                                                   |
|--------|-------------------------------------------------------------------------|
| parent | —                                                                       |
| kind   | task · type Task                                                        |
| title  | Maintain the gate decision log and the narrated five-minute demo script |
| gate   | G1–G6                                                                   |

| Field       | Value                                                                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| description | The decision log (§13) records G1-G6 outcomes + the edge call; the demo script narrates: plain-UDP slice → block WG → MASQUE survives → flip a map entry → new network reachable [04_P0 §11.7-8, §12]. Decision log is the authoritative Phase-0→Phase-1 entry record. |
| depends_on  | HVPN-P0-048, HVPN-P0-070, HVPN-P0-074                                                                                                                                                                                                                                  |
| deliverable | docs/spike/decision_log.md + docs/spike/demo.md + recording                                                                                                                                                                                                            |
| acceptance  | all six gate rows filled with evidence paths; demo recording validated (§11.4.163); gates table all non-pending                                                                                                                                                        |
| est_effort  | 0.5                                                                                                                                                                                                                                                                    |
| test_types  | CH, HQA, FA                                                                                                                                                                                                                                                            |

### **TASK HVPN-P0-079 — Workable-items DB seed + docs\_chain sync**

| Field       | Value                                                                                                                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent      | —                                                                                                                                                                                                                                   |
| kind        | task · type Task                                                                                                                                                                                                                    |
| title       | Seed the workable-items SQLite DB from this WBS and wire docs_chain sync                                                                                                                                                            |
| gate        | —                                                                                                                                                                                                                                   |
| description | This WBS is the single source for docs/workable_items.db (§11.4.93/.95). Seed every HVPN-P0-NNN row + the six gates; wire docs_chain (§11.4.106) so md↔DB stay byte-identical (no force-push, §11.4.113; git-tracked DB, §11.4.95). |
| depends_on  | HVPN-P0-002                                                                                                                                                                                                                         |
| deliverable | seeded docs/workable_items.db + .docs_chain/contexts/phase0.yaml                                                                                                                                                                    |
| acceptance  | workable-items validate passes; md→db→md round-trip byte-identical; DB committed (tracked)                                                                                                                                          |
| est_effort  | 0.5                                                                                                                                                                                                                                 |
| test_types  | UT, IT, FA                                                                                                                                                                                                                          |

### **TASK HVPN-P0-080 — make spike one-shot + containers-submodule integration infra**

| Field  | Value |
|--------|-------|
| parent | —     |

| Field       | Value                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kind        | task · type Task                                                                                                                                                                                                                                                                                                              |
| title       | Wrap the whole rig in make spike on rootless-Podman integration infra                                                                                                                                                                                                                                                         |
| gate        | G1, G2                                                                                                                                                                                                                                                                                                                        |
| description | The reproducible rig + DPI-block + netem scripts run in one make spike [04_P0 §11.3]; integration infra (LAN sim hosts, edge containers) boots <b>on-demand via the containers submodule</b> (§11.4.76) in <b>rootless Podman</b> (§11.4.161) — no ad-hoc docker. Quadlet manifests describe the edge + connector containers. |
| depends_on  | HVPN-P0-022, HVPN-P0-040, HVPN-P0-043                                                                                                                                                                                                                                                                                         |
| deliverable | Makefile (make spike) + deploy/spike/*.container quadlets                                                                                                                                                                                                                                                                     |
| acceptance  | make spike runs S0-S2 + S3 slice headless, deterministic over N=3 (§11.4.50); infra booted via containers submodule; captured run log                                                                                                                                                                                         |
| est_effort  | 1.0                                                                                                                                                                                                                                                                                                                           |
| test_types  | IT, FA, SC                                                                                                                                                                                                                                                                                                                    |

Rootless-Podman quadlet for the spike edge (boots on-demand, §11.4.76/.161):

```
# deploy/spike/helix-edge.container — rootless Podman quadlet
[Unit]
Description=HelixVPN Phase-0 edge (MASQUE termination)
After=network-online.target

[Container]
Image=localhost/helix-edge:spike
PublishPort=443:443/udp
AddCapability=NET_ADMIN
ReadOnly=true
NoNewPrivileges=true
# rootless: no root, no sudo, no SSH on edge (§7 security invariants)

[Service]
Restart=on-failure

[Install]
WantedBy=default.target
```

## 13. Decision log (fill as gates clear)

### [04\_P0 §12]

| Gate                               | Outcome                                                     | Evidence (numbers / capture)                                              | Decision / consequence                          | Items            |
|------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------|------------------|
| <b>G1</b> plain-UDP slice          | <input type="checkbox"/> pass <input type="checkbox"/> fail | iperf ... ,<br>ping/curl ...                                              | —                                               | HVPN-P0-025/-027 |
| <b>G2</b> MASQUE through DPI block | <input type="checkbox"/> pass <input type="checkbox"/> fail | goodput@5%<br>loss ... ,<br>tshark ...                                    | obfuscation<br>D1 confirmed                     | HVPN-P0-035/-038 |
| <b>G3</b> iOS memory               | <input type="checkbox"/> pass <input type="checkbox"/> fail | peak RSS<br>... / ceiling<br>... /<br>headroom ...<br>%                   | core stays<br>Rust ↔ §6.4<br>fallback<br>ladder | HVPN-P0-067/-070 |
| <b>G4</b> edge language            | <input type="checkbox"/> Rust <input type="checkbox"/> Go   | CPU/Gbps<br>R:... G:... ;<br>p99 R:...<br>G:... ;<br>MASQUE<br>effort ... | single-impl ↔<br>dual-impl<br>(D5)              | HVPN-P0-045/-048 |
| <b>G5</b> FFI                      | <input type="checkbox"/> pass <input type="checkbox"/> fail | demo<br>recording                                                         | frb v2 ↔<br>UniFFI                              | HVPN-P0-052/-054 |
| <b>G6</b> reconcile                | <input type="checkbox"/> pass <input type="checkbox"/> fail | map-delta →<br>reachable,<br>no restart                                   | confirms push<br>model                          | HVPN-P0-074/-076 |

## 14. Phase-0 risks + pre-planned fallbacks [04\_P0 §13]

| Risk                                   | Likelihood | Fallback                                                                | Item(s)          | Decision re-opened (reversal criteria)                                     |
|----------------------------------------|------------|-------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------|
| MASQUE/CONNECT-UDP in Rust over budget | med-high   | Prototype MASQUE first in Go (masque-go) to unblock G2, then port — and | HVPN-P0-031/-042 | <b>D5</b> (edge language) — reversal fires if this pushes G4 toward Go per |

| Risk                                | Likelihood | Fallback                                                                                                                                       | Item(s)     | Decision re-opened (reversal criteria)                                                                                                         |
|-------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| boringtun quirks / maintenance gaps | med        | let that inform G4 toward Go<br>Swap to wireguard-go via cgo for the spike, or kernel WG on Linux for non-iOS gates                            | HVPN-P0-011 | decision-register.md §2 D5<br><br>none registered (implementation swap, not a D-decision)                                                      |
| iOS NE memory fails                 | med        | §6.4 ladder; worst case iOS ships plain-WG + on-path obfs only in v1                                                                           | HVPN-P0-067 | <b>D2</b> (Rust core, make-or-break) — a captured G3 fail re-opens D2 and the whole client strategy per decision-register.md §2 D2 / §6        |
| h3 immaturity blocks datagrams      | med        | Pin versions; if blocked, carry WG over a QUIC <i>stream</i> as a stopgap (accept HoL blocking) purely to pass G2, flag non-final              | HVPN-P0-028 | <b>D1</b> (MASQUE primary) — a stopgap-stream G2 pass is provisional; genuine reversal needs a captured G2 fail per decision-register.md §2 D1 |
| Time-box blown                      | —          | Stop; the unfinished gate <b>is</b> the finding — escalate the architectural decision with partial data (§11.4.6) rather than overrun silently | HVPN-P0-078 | whichever gate's decision (D1/D2/D5) is left unresolved stays a recommendation, never a resolution, per decision-register.md §0                |

**Traceability note.** This risk register names the *operational* fallback per item; v00-meta/decision-register.md §1/§2/§6 is the *canonical* record of which program-level decision (D1/D2/D5) each Phase-0 gate resolves and the exact captured-evidence condition that would re-open it. The two documents are deliberately non-duplicative: this table is "what do we do Monday morning if a gate fails," the decision register is "which cross-cutting architectural call does that failure re-open, and under what evidence."

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## 15. How Phase 0 feeds Phase 1 [04\_P0 §14]

Every surviving interface graduates: the Transport trait (HVPN-P0-004) gains Shadowsocks/UoT/LWO impls; the static map.json shape (HVPN-P0-071) becomes the WatchNetworkMap protobuf streamed by the Go coordinator [02-control-plane, 03-protocol]; the FFI surface (HVPN-P0-049) gains policy/multi-hop fields [03-client-core-and-ui]; the winning edge (G4) becomes helix-edge in the monorepo [04\_ARCH §11]; the bench.sh harness (HVPN-P0-046) becomes the CI-less local gate suite with the §8 bars as thresholds. Nothing built here is wasted **if the interfaces hold** — which is the entire discipline of Phase 0.

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*Constitution bindings applied: §11.4.44 (revision header), §11.4.54 (stable monotonic HVPN-P0-NNN ids), §11.4.91 (≥6-word self-contained titles), §11.4.93/.95 (this WBS is the git-tracked workable-items DB source; DDL in §0.3), §11.4.106 (docs\_chain md↔DB sync), §11.4.169 (per-item test\_types from the closed set; §0.4), §11.4.5/.69/.107 (every acceptance cites captured PHYSICAL evidence), §11.4.6 (no-guessing — measure, don't assume; gate fail = finding), §11.4.50 (deterministic N=3), §11.4.76/.161 (containers submodule + rootless Podman for integration infra), §11.4.119 (bench rig = single-owner exclusive resource), §11.4.144 (device-availability followed during device runs), §11.4.159/.163 (window-scoped validated recordings), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement).*